

Paraneoplastic Neurological Syndromes: Unusual Presentations of Cancer. A Practical Review

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Abstract: *Introduction:* Paraneoplastic neurological syndromes (PNS) are uncommon and imperfectly understood and, therefore, are frequently underdiagnosed. *Methods:* We review the current literature on the clinical presentation, diagnosis, pathophysiology and treatment of PNS. *Results:* PNS are a heterogeneous group of neurologic disorders caused by an immune response to an underlying malignancy. A 3-step diagnostic process is necessary to establish the diagnosis. We review the role of onconeural antibodies in the diagnosis and pathogenesis of PNS and describe recent advances in treatment, focusing on paraneoplastic encephalomyelitis, limbic encephalitis, paraneoplastic cerebellar degeneration, opsoclonus-myoclonus syndrome, subacute sensory neuropathy and Lambert-Eaton myasthenic syndrome. *Conclusion:* PNS often antedate the diagnosis of cancer, offering an opportunity for detecting cancer at an earlier and curable stage. Tests for paraneoplastic antibodies are often negative and do not rule out the diagnosis of a paraneoplastic syndrome. Certain clinical presentations should suggest a paraneoplastic syndrome, even in the absence of paraneoplastic antibodies, and prompt a thorough search for occult malignancy.

Key Indexing Terms: Paraneoplastic; Nervous system; Neoplasm; Autoantibodies. [Am J Med Sci 2010;340(4):301–308.]

DEFINITION

Paraneoplastic neurological syndromes (PNS) are a heterogeneous group of neurologic disorders caused by an immune response to an underlying malignancy. Thus, the neurologic effects are not the result of metastases, nutritional abnormalities, amyloid deposition or adverse effects of cancer treatments.

INCIDENCE

The exact incidence of PNS among those diagnosed with cancer is still uncertain, with estimates ranging from 1 in 10,000 to 1 in 100. The neurologic syndrome precedes the diagnosis of cancer in 50% to 80% of cases.¹

The reported incidence of specific syndromes varies. Paraneoplastic sensory neuropathy (SN) is probably the most common (3–7 per 1000 cancer diagnoses),² followed closely by paraneoplastic encephalitis (3 per 1000) and cerebellar degeneration (2 per 1000). However, most estimates are from referral centers and not from population-based studies. Another source of error that precludes an accurate estimate of the incidence of PNS is the frequent use of an overly restrictive case definition

that requires positive paraneoplastic antibodies. The lack of a unifying definition for PNS also contributes to the problem.

PATHOGENESIS

A cancer-stimulated immune reaction that cross-reacts with neural tissue—onconeural immunity—is considered the principal pathologic mechanism for PNS. Some cancer cells express proteins that are normally restricted to the nervous system. An immune response that targets these proteins can then cross-react with noncancerous cells that express the same proteins. For example, when serum from a patient with limbic encephalitis, one of the PNS, was incubated with the patient's cancer cells and with tissue from a rat's brain, there was evidence for antibody fixation to the same Ma proteins on both neurons and cancer cells.³

To date, there is no evidence that antineural antibodies associated with paraneoplastic syndromes of the central nervous system (CNS) are pathogenic, although they serve as important markers for paraneoplasia. In some PNS, circumstantial evidence suggests that T cell-mediated mechanisms play a major pathogenic role.⁴

Finally, not all patients with PNS and detectable onconeural antibodies will have a cancer diagnosed. In a series of 71 patients with PNS and anti-Hu antibodies, no cancer was detected in 13%.^{5,6} It is not known whether this represents successful immune tumor suppression or whether the antibodies were stimulated by a nonmalignant process.

DIAGNOSIS

PNS offers an opportunity for diagnosing cancer at an early stage because, for 50% to 80% of patients, the PNS presents before there is any other indication of malignancy.^{7,8}

The diagnosis of PNS is a 3-step process. First, other etiologies for the neurological syndrome must be considered and ruled out. Second, there must be an aggressive search for an underlying malignancy. And third, the diagnosis of PNS must be confirmed using a combination of clinical features, laboratory data and patient follow-up (Figure 1).

Step 1: Rule Out Other Diagnoses

A number of other etiologies can cause similar neurological syndromes, including vascular insults, infections, connective tissue diseases, nutritional deficiencies, adverse effects of medications, toxic exposures and metabolic derangements. A careful history and physical examination will often provide helpful clues, but laboratory and imaging tests will usually be necessary. We suggest starting the evaluation with a complete blood count (looking for the macrocytosis or the multilobed PMNs associated with vitamin B12 deficiency), a basic metabolic panel, liver function tests, chronic hepatitis profile and antinuclear antibody (ANA) test.

Examination of the cerebrospinal fluid (CSF) can help exclude infection or leptomeningeal spread of cancer. Some PNS will have suggestive but nonspecific findings on CSF

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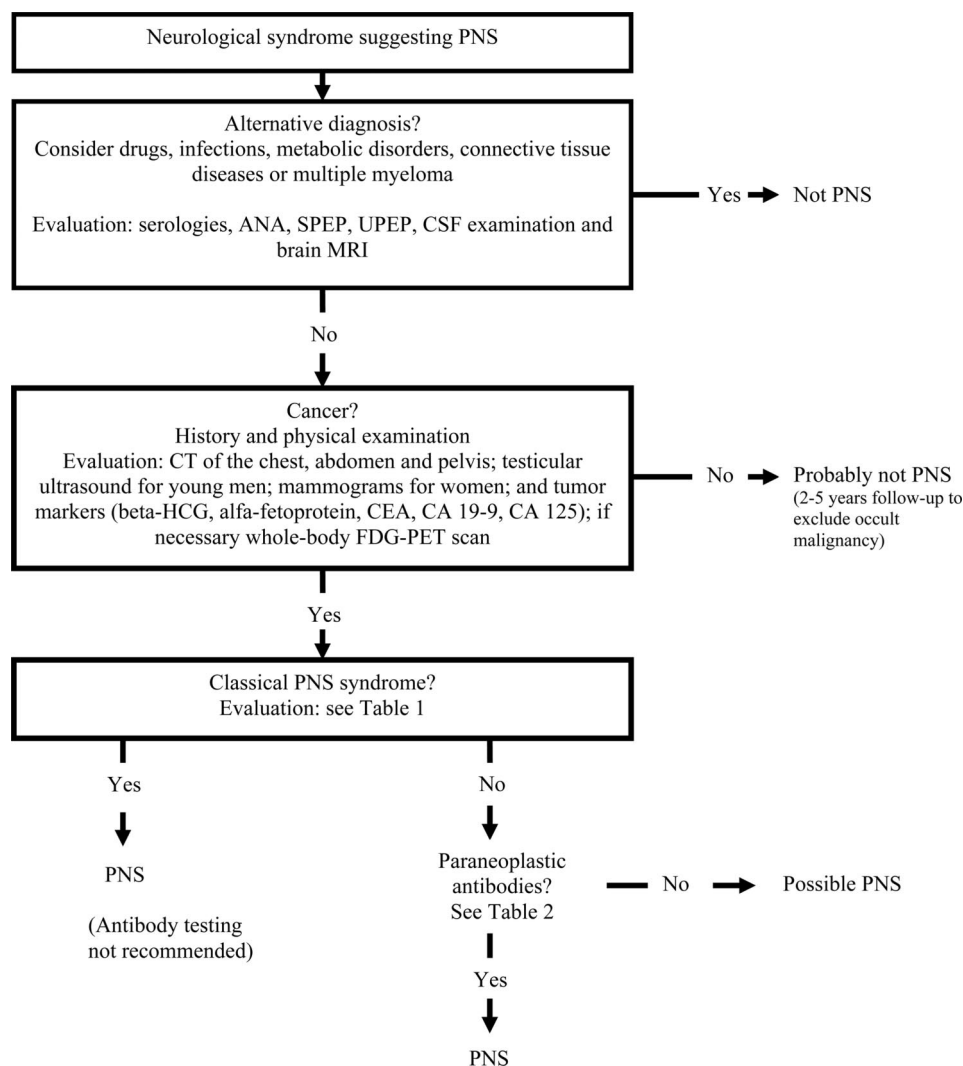


FIGURE 1. Diagnostic evaluation of patients suspected of having paraneoplastic neurological syndromes.

examination. Magnetic resonance imaging (MRI) is important in evaluating for primary CNS tumors and metastatic involvement, vascular disease, multiple sclerosis and, for certain PNS, such as limbic encephalitis.

Step 2: Search for a Hidden Malignancy

After excluding most of the competing etiologies, the next step is a thorough search for malignancy. Many clinicians will order tests to look for paraneoplastic antibodies before looking for cancer. This approach is not recommended, because a negative test for paraneoplastic antibodies might dissuade the clinician from an appropriately aggressive search for cancer or there might be an inappropriate delay in cancer diagnosis while awaiting antibody test results.

Usually, a thorough review of systems and physical examination will help focus the search for malignancy. Also, certain neurologic syndromes are associated with specific malignancies; for example, Lambert-Eaton myasthenic syndrome (LEMS) is most often associated with small-cell lung cancer.

If the initial focused search is unrevealing, then it is usually necessary to perform computed tomography scans of the chest, abdomen and pelvis. If these tests are negative or

equivocal for cancer, then the final step is usually a whole body fluoro-2-deoxy-glucose (FDG) positron emission tomography (PET) scan, which is not routinely available at many hospitals.^{6,9}

In a small study of 13 patients with PNS and positive paraneoplastic antibodies, the sensitivity of computed tomography scanning for detecting malignancy was 30%, whereas the sensitivity of FDG-PET scanning was significantly higher, 90% ($P < 0.01$). Although the sensitivity of the combination of both methods was 100%, because of the small number of patients, the confidence interval was broad (95% CI: 75%–100%).⁶ Gonadal tumors might evade detection by PET scanning because some do not take up the FDG. Intratubular germ cell neoplasias can also be undetectable to all tests, at least at early stages, and require careful follow-up and repeat focused testing.¹⁰

The CSF of patients with PNS is usually nonspecific: moderate lymphocytic pleocytosis, increased protein concentration, high IgG index and CSF-specific oligoclonal bands. When patients develop paraneoplastic antibodies, they are usually detectable in both the serum and CSF. In some instances, the titers of the autoantibodies might be

TABLE 1. Paraneoplastic neurological syndromes

Classical syndromes	Nonclassical syndromes
Encephalomyelitis	Brainstem encephalitis
Limbic encephalitis	Optic neuritis
Subacute cerebellar degeneration	Cancer-associated retinopathy
Opsoclonus-myoclonus	Melanoma-associated retinopathy
Subacute sensory neuropathy	Stiff person syndrome
Lambert-Eaton myasthenic syndrome	Necrotizing myelopathy
	Motor neuron diseases
	Acute sensorimotor neuropathy
	Guillain-Barre syndrome
	Brachial neuritis
	Subacute/chronic sensorimotor neuropathies
	Neuropathy and paraproteinemia
	Neuropathy with vasculitis

Table modified from Graus F, Delattre JY, Antoine JC, et al. Recommended diagnostic criteria for paraneoplastic neurological syndromes. *J Neurol Neurosurg Psychiatry* 2004;75:1135–40. Copyright © 1995 National Academy of Sciences USA.

higher in the CSF than in the serum, which explains why some disorders, such as anti-*N*-methyl-D-aspartate receptor (anti-NMDAR) encephalitis, have a CSF positive for antibodies whereas the serum is negative. This finding has been attributed by Vega et al¹¹ to intrathecal paraneoplastic antibody synthesis rather than passive transfer of blood-derived antibodies to the CSF.

As mentioned earlier, MRI is useful because it can rule out metastatic complications of cancer and because it can also suggest the diagnosis of limbic encephalitis, with increased signal in the medial portion of one or both temporal lobes on

FLAIR or T2 sequences, although the findings are not definitive.^{12,13} Limbic encephalitis can also cause foci of epileptic activity in one or both temporal lobes on electroencephalogram testing,¹³ but again, these findings are nonspecific.

Step 3: Establish the Diagnosis of PNS

To help clinicians differentiate between true PNS from neurologic symptoms that are coincidental with a cancer, an international panel of neurologists proposed criteria to establish a “definite” or “possible” diagnosis of PNS.¹⁴

After excluding other possible causes of the neurologic syndrome (discussed in step 1 above), a “definite” diagnosis of PNS is made if the following 2 criteria are satisfied: (1) a classic syndrome (ie, one of the syndromes frequently associated with cancer; Table 1); and (2) cancer diagnosed within 5 years of the neurologic syndrome. For patients meeting these criteria, a diagnosis of PNS can be established even if the search for antibodies was negative. In the setting of a nonclassic neurologic syndrome, a positive antibody test and a cancer diagnosis satisfy criteria for a definite PNS diagnosis (assuming the step 1 evaluation was unrevealing).

AUTOANTIBODIES

Although PNS is an immune-mediated process, onconeural antibodies are not always detected. There are 2 likely reasons: first, many onconeural antibodies have not yet been discovered, and, second, some PNS are best explained by nonhumoral immune mechanisms, such as cytotoxic T cells. Specific paraneoplastic autoantibodies can be associated with several different neurologic syndromes, but typically, they are highly predictive of particular cancers (Table 2).

The majority of paraneoplastic antibodies are directed against intracellular antigens in the nucleus or cytoplasm of neurons. In some cases, the protein antigen has been identified definitively, and testing using western blot against recombinant protein is available. In other cases, the antibodies are defined descriptively based on the pattern of immunohistochemical staining of brain sections.¹⁵

TABLE 2. Paraneoplastic autoantibodies and associated cancers and syndromes

Autoantibody	Associated cancer	Paraneoplastic neurologic syndrome
Well-recognized antibodies		
Anti-Hu	Small cell lung cancer	Encephalomyelitis, paraneoplastic cerebellar degeneration, sensory neuronopathy
Anti-Yo	Gynecologic and breast cancer	Paraneoplastic cerebellar degeneration
Anti-Ri	Breast, gynecologic and small cell lung cancer	Paraneoplastic cerebellar degeneration, opsoclonus-myoclonus
Anti-Tr		Paraneoplastic cerebellar degeneration
Anti-CV2 or anti-CRMP5	Hodgkin lymphoma	Encephalomyelitis, paraneoplastic cerebellar degeneration, sensory neuronopathy
Anti-Ma proteins	Small cell lung cancer	Limbic encephalitis, paraneoplastic cerebellar degeneration
Antiamphiphysin	Germ-cell tumors of testis, breast cancer	Stiff-man syndrome, encephalomyelitis
Partially recognized antibodies		
Antivariable-gated calcium channel	Small cell lung cancer	Lambert-Eaton myasthenic syndrome, paraneoplastic cerebellar degeneration
Antiacetylcholine receptor	Thymoma	Myasthenia gravis
Antivariable-gated potassium channel	Thymoma	Limbic encephalitis

Table modified from Graus F, Dalmau J. Paraneoplastic neurological syndromes: diagnosis and treatment. *Curr Opin Neurol* 2007;20:732–7. Copyright © 1995 National Academy of Sciences USA.

Expert consensus has classified some autoantibodies as “well-recognized” paraneoplastic antibodies based on the weight of the scientific evidence; whereas others are designated as “partially recognized” because the current scientific support is weaker.¹⁴ The best characterized paraneoplastic antibodies (and their antigenic targets) include the following: anti-Hu [antineuronal nuclear antibody (ANNA) 1], anti-Yo [Purkinje cell cytoplasmic antibody type (PCA) 1], anti-CV2 (collapsin-response mediator protein 5), anti-Ri (ANNA2), anti-Ma2 (Ta) and antiampiphysin antibodies. Because these antibodies are so highly specific for cancer, it is difficult to exclude the diagnosis of cancer in the presence of one of these antibodies without an exhaustive evaluation and a prolonged period of clinical follow-up.¹⁶ Although antivoltage-gated calcium channel (anti-VGCC) antibodies clearly cause the LEMS, the anti-VGCC antibodies are not included in the list above because they are not specific for cancer. Perhaps, 40% of patients with this syndrome have an autoimmune disease unassociated with cancer.

Several other antibodies have been proposed as possible paraneoplastic autoantibodies, but until future research establishes their etiologic role, they will continue to be classified as “partially recognized”: anti-Tr (PCA-Tr), ANNA3, PCA2, anti-Zic4 and anti-mGluR1.

It is recommended that to confirm the presence of onconeural antibodies, the immunohistochemistry and western blot tests must both be positive. Neither test alone is reliable because of cross-reactivity with other autoantibodies, such as those associated with Sjogren’s syndrome, which also includes autoimmune neuropathies.^{17,19} Finally, in studies thus far, serial measurements of paraneoplastic antibodies have not been proved clinically useful.²⁰

CLINICAL PRESENTATION: SPECIFIC NEUROLOGIC SYNDROMES

Some PNS have been classified as “classical” because they are most often associated with cancer (Table 1). In the following section, we will focus on the clinical presentations of the classical PNS: paraneoplastic encephalomyelitis (PEM), limbic encephalitis, paraneoplastic cerebellar degeneration (PCD), opsoclonus-myoclonus syndrome (OMS), subacute sensory neuropathy and LEMS. Physicians should realize that other “nonclassical” PNS do exist, such as paraneoplastic stiff-man syndrome; motor neuron disease; paraneoplastic affliction of nerve and muscle secondary to vasculitis; acute necrotizing myopathy; polymyositis and dermatomyositis; autonomic dysfunction; neuromyotonia and sensorimotor neuropathy associated with multiple myeloma.

Paraneoplastic Encephalomyelitis

This disorder is characterized by multiple neurologic symptoms as a result of an inflammatory disorder that may involve any part of the CNS, dorsal root ganglia and autonomic nerves.³

Small-cell lung carcinoma is the most common malignant tumor associated with this disorder, but lymphoma, carcinoma of the esophagus, breast, ovaries, pancreas and kidney, germ cell tumors and hepatocellular carcinoma have also been reported.^{5,8,21–26} The neurologic symptoms usually precede the diagnosis of cancer. In most cases, cancer is detected within 4 to 12 months after the onset of neurologic symptoms, but in some cases, cancer has not been diagnosed until 8 years later.^{5,23}

Wilkinson and Zeromski²⁷ were the first to identify the presence of antineuronal nuclear antibodies in patients with this disorder. The polyclonal IgG antibodies are directed against a

35- to 40-kD protein or complex of proteins, the so-called Hu antigen (named after the first 2 letters of the name of one of the patients with the disorder). The detection of anti-Hu antibodies in serum is correlated with the presence of sensory neuropathy, whereas the detection of antibodies in the CSF is associated with the presence of PEM.¹¹ The reported sensitivity of a positive test for anti-Hu antibodies is 88%, and the specificity is 99%.²⁸ The Hu antigen is a family of 4 similar RNA-binding proteins (HuD, HuC/ple21, Hel-N1 and Hel-N2). The antigens are expressed in the nuclei and, to a lesser extent, in the cytoplasm of neurons and cells of small-cell lung carcinoma.²⁹

In a European retrospective analysis of 200 patients with Hu-positive PEM, the following clinical characteristics were observed²⁴:

Neurologic dysfunction was confined to 1 area of the nervous system in 60 patients (30%) [SN (48), cerebellar ataxia (4), limbic encephalitis (4), brainstem encephalitis (2), intestinal pseudo-obstruction (1) and parietal encephalitis (1)]. The other patients had evidence of multifocal involvement. SN was the most frequent predominant syndrome at diagnosis in 108 (54%) patients. The clinical features were those of large-fiber SN. An asymmetrical SN, sometimes resembling mononeuritis multiplex, is the most common presentation. The sensory symptoms in PEM patients are usually because of a lesion in the dorsal root ganglia rather than in the peripheral nerve, as occurs in the common sensorimotor neuropathies.³⁰ Regardless of the presentation, the disorder usually evolves in weeks to months to a widespread encephalomyelopathy with sensory and autonomic deficits.³¹ An acute respiratory or cardiac dysautonomia leading to sudden cardiac death has been reported.⁵ Some patients with anti-Hu PEM/SN demonstrate focal neurologic disorder throughout the course of their clinical illness, as had been reported by Shavit et al,³² who reported 3 cases of anti-Hu PEM presenting as “epilepsia partialis continua,” a unique type of prolonged seizure, in which patients experience recurrent motor epileptic seizures that are focal and recur every few seconds or minutes for extended periods (days or years). Sensorimotor neuropathy is also seen accompanying PEM mediated by anti-CV2 and anti- α -enolase antibodies.³³ In these cases, patients present with progressive muscle weakness of the extremities associated with absence of deep tendon reflexes and stocking distribution of dysesthesias. This form of neuropathy has features of axonal and myelin damage.

Besides anti-Hu antibody, other antineuronal antibodies have been associated with PEM, such as antiampiphysin,³⁴ anti-Ma in patients with testicular cancer,²⁴ anti-CV2 in patients with thymoma,³⁵ anti- α -enolase in patients with gastric adenocarcinoma,³³ anti-NMDAR antibodies in patients with teratomas³⁶ and anti-GAD in patients with pancreatic tumors.²⁵

Anti-NMDAR encephalitis offers an example of paraneoplastic syndrome improvement with treatment of underlying tumor. It has recently been related to the development of antibodies to the NR1/NR2B heteromers of the NMDAR.³⁷ This disorder results in a characteristic syndrome in young women with teratoma. It presents with prominent psychiatric symptoms or, less frequently, memory deficits, followed by a rapid decline of the level of consciousness, central hypoventilation, seizures, involuntary movements and dysautonomia. Despite the severity of symptoms and prolonged clinical course, most patients recover if the disorder is recognized and treated.³⁸ Anti-NMDAR antibodies bind to the extracellular conformational epitope in NR1/NR2 heteromers of the NMDAR, although recent data have demonstrated that its critical epitope region resides in the NR1 subunit. CSF reveals nonspecific changes. Electroencephalogram usually shows diffuse

slowing without paroxysmal discharges.³⁹ A tumor was found in 58% of patients with anti-NMDAR encephalitis.⁴⁰

Treatment of the underlying cancer generally does not affect the course of PEM with sensory neuropathy.⁵ However, the condition may improve or stabilize with the treatment of the tumor.^{24,41,42} Unfortunately, treatment with plasmapheresis, intravenous immunoglobulin (IVIG) and immunosuppressive agents has had disappointing results.^{5,23,43} Nevertheless, a few patients may benefit from immunotherapy.⁸ In rare instances, PEM improves spontaneously.³²

Anti-NMDAR encephalitis offers an exception to the rule that treatment of the underlying tumor does not affect the course of the neurologic symptoms. Approximately 65% of patients with anti-NMDAR encephalitis had full or near-full recovery.⁴⁰ Treatment of anti-NMDAR encephalitis associated with teratoma includes immunotherapy and/or tumor removal. Corticosteroids and IVIG are effective in patients with and without tumor. It is likely that patients who do not respond to one form of immunotherapy might respond to other regimens including plasmapheresis, cyclophosphamide and rituximab. Early removal of tumor should be considered based on the following reasons. First, patients with ovarian teratoma showed higher mortality and higher titer of anti-NMDA antibody compared with those without. Second, relapsing neurologic symptoms occurred in 13% of patients, usually related to a delay in tumor diagnosis. Third, when a tumor was found and removed, recovery was faster and predictable. However, early removal of tumor cannot always be conducted because of unstable conditions such as hypoventilation and dyskinesias.⁴⁰

Limbic Encephalitis

The cardinal sign of limbic encephalitis is a severe impairment of short-term memory. It classically presents with rapid development of irritability, depression and partial complex or generalized seizures. Most patients seem confused. The subacute onset of short-term memory deficits is considered the hallmark of the disorder, although it is often overlooked because of patients' confusion or because of other symptoms that overshadow the memory loss.⁴⁴ It is pathologically characterized by an inflammatory process confined to the limbic system. In most cases, diagnosis is suggested by the clinical picture in combination with findings on MRI and electrophysiologic studies, as described in the diagnosis section.

Limbic encephalitis is associated mainly with anti-Hu and anti-Ma2 antibodies and, less frequently, with anti-CV2 and antiampiphysin antibodies. Tumors commonly associated with limbic encephalitis are small-cell lung cancer (in patients with anti-Hu antibodies) and thymoma. In men younger than the age of 50 years with anti-Ma2 antibodies, limbic encephalitis is almost always associated with testicular germ cell tumors, which can be microscopic and difficult to detect.

Paraneoplastic Cerebellar Degeneration

Although PCD is one of the better-known syndromes, it is rare. Neurologic deficits are sometimes preceded by prodromal symptoms, such as a viral-like illness, dizziness, nausea or vomiting, which might be attributed to a peripheral vestibular process. These symptoms are followed by gait unsteadiness that rapidly develops into ataxia, diplopia, dysarthria and dysphagia. Some patients have blurry vision, oscillopsia and transient opsoclonus.⁷ Cranial MRI in PCD varies according to the timing of the study in relation to the course of illness. It is usually normal in the initial stages. Sometimes, MRI shows swelling in the cerebellar hemispheres and increased signal activity. Later in the course of the illness, MRI usually shows

cerebellar atrophy.⁴⁵ The main pathologic feature of PCD is an extensive loss of Purkinje cells that might be associated with inflammatory infiltrates in the cerebellar cortex, deep cerebellar nuclei and inferior olivary nuclei.^{46,47}

PCD is commonly associated with anti-Yo antibodies in women with breast or ovarian malignancies,⁴⁸ but it is also seen in patients with Hodgkin lymphoma associated with anti-Tr antibodies.⁴⁹ Patients with small-cell lung cancer can develop one or multiple immune responses in association with PCD. In this setting, up to 41% of patients develop antibodies against VGCCs with or without associated LEMS, 23% develop anti-Hu antibodies and a minority develops other antibodies, such as antibodies against collapsin-response mediator protein 5 (CRMP5 or CV2), amphiphysin, PCA2 or ANNA3.⁷

The cerebellar syndrome typically progresses over weeks to months and then stabilizes. Patients, however, may be so debilitated by this syndrome that they require intensive supportive care. Treatment options have not been promising so far. Several modalities of immunotherapy have been tried including steroids, plasmapheresis, IVIG, tacrolimus and rituximab.⁵⁰ One series showed an impressive 40% response to plasmapheresis.⁵¹ However, this was not reproduced in any of 16 patients treated in another report.⁴³ IVIG has been reported to be effective in patients even with longstanding disease.^{52,53} Finally, some patients have noted an improvement with treatment of the primary tumor, but, usually, only if complete eradication can be achieved.

Opsoclonus-Myoclonus Syndrome

Opsoclonus is a dyskinesia consisting of involuntary, arrhythmic, chaotic, multidirectional saccades with horizontal, vertical and torsional components and is commonly accompanied by myoclonic jerks in the limbs and trunk, cerebellar ataxia, tremor and encephalopathy.⁵⁴ Such saccadic oscillations in opsoclonus differ from nystagmus in that the phase that takes the eye off the target is always a saccade, not a smooth eye movement. In children, sleep disturbances, including prolonged sleep latency, fragmented sleep, reduced quantity of sleep, snoring and nonrestorative sleep, were reported in addition to frequent rage attacks.⁵⁵ The median age of children presenting with this disorder is about 18 months. OMS has been reported in association with cancer, in viral infections, in exposure to toxic metabolites or as idiopathic cases.⁵⁵

The etiology of OMS is still unclear, but the presence of autoantibodies, described in both adult and pediatric OMS, points to an autoimmune origin. In this regard, anti-Hu antibodies and a specific antineuronal antibody (anti-Ri) were identified in some adult patients affected by OMS associated with small-cell lung and breast cancers, respectively.^{56,57} In children, the disorder is related to the presence of a neuroblastoma in approximately 50% of cases, in which serum autoantibodies against neurons and cerebellar Purkinje cells have been consistently detected. The specificity of these antibodies was limited, and the target molecules have not been identified yet.⁵⁸

Although nearly all the well-characterized paraneoplastic antibodies have been reported in single case reports, most patients (both children and adults) are antibody negative.⁷ Paraneoplastic OMS is less responsive to immunotherapy. Corticosteroids or IVIG might speed up improvement in patients with idiopathic opsoclonus, but not in those with paraneoplastic opsoclonus; the latter only responded when the tumor was controlled.^{7,59} Immunotherapy seems to be helpful, but improvement is mild or not sustained unless the tumor is controlled.⁶⁰

Subacute SN

This is the most common PNS, although it is probably underdiagnosed. Most cases present with symmetric or asymmetric paresthesias and hypoesthesias affecting the extremities, trunk and face. Vibration and joint position may be affected disproportionately to other modalities of sensation, hence, patient sometimes can present with sensory ataxia. A recent study found that only 4 of 14 patients with a clinical SN had an isolated axonal/neuronal electrophysiologic pattern of pure sensory nerve involvement.⁶¹ Pain was prominent in 85% of cases.⁶¹ The course of the neuropathy was acute (5%), progressive (40%) or subacute (55%). Although the clinical picture is purely sensory, electrophysiologic studies may demonstrate some motor involvement. The typical electrophysiologic findings of dorsal root ganglionitis include decreased or absent sensory nerve potentials with normal or near normal motor conduction velocities.

Symptoms result from a dorsal root ganglionitis characterized by inflammatory infiltrates, neuronal degeneration, proliferation of satellite cells and secondary Wallerian degeneration of the spinal cord.⁶ This syndrome is often associated with small-cell lung cancer and with anti-Hu or anti-CV2 antibodies.²⁸

Paraneoplastic SN rarely improves with treatment of the tumor.⁶ Although some patients treated with IVIG may show stabilization of symptoms or mild improvement, the roles of this treatment and plasma exchange are not proven.³

Lambert-Eaton Myasthenic Syndrome

Patients with LEMS often present with proximal leg weakness causing problems during walking or climbing stairs. Symptoms of autonomic dysfunction—dry mouth and constipation—are also common. Reflexes are usually reduced but can improve after exercise. Antibodies against presynaptic VGCCs cause less acetylcholine to be released with every nerve stimulation, affecting the function of motor nerves, postganglionic sympathetic nerves and parasympathetic nerves.

On electrophysiologic testing, patients with LEMS have decreased compound muscle action potential amplitudes, often 10% of normal, which decrease further with repetitive nerve stimulation between 1 and 5 Hz. Motor and sensory latencies and conduction velocities are normal. After exercise, or with repetitive nerve stimulation at 20 Hz or more for up to 10 seconds, compound muscle action potential amplitudes increase 100% or more, a phenomenon resulting from buildup of calcium within the nerve terminal more rapidly than it can be removed by mitochondria, allowing for increased acetylcholine (Ach) quantal release.⁶²

Pathophysiology is based on P/Q VGCC antibody, present in almost 100% of patients with cancer but 90% without cancer.^{62,63} The channel resides on the presynaptic nerve terminal and, when depolarized, allows calcium to enter the cell. Calcium is required for the quantal release of Ach from the presynaptic nerve terminal. Ach then crosses the neuromuscular junction, binds to the postsynaptic Ach receptors on the muscle membrane and results in muscle depolarization and contraction. P/Q VGCC antibody exerts its affect by cross-linking and downregulating the VGCC, preventing calcium influx into cells and Ach release.⁶²

Diagnosis is based on the classic electromyographic findings and on finding of anti-VGCC serum antibodies (sensitivity 90%). Although the majority of patients with LEMS has a malignancy (usually small-cell lung cancer), some patients will not have a detectable malignancy but may have other autoimmune diseases.

Treatment of paraneoplastic LEMS with immunotherapy without treatment of the underlying malignancy showed disappointing results.⁶⁴ 3,4-Diaminopyridine, a drug that was used initially in Duke university, showed a significant improvement in LEMS symptoms in a randomized trial reported by Sanders et al.⁶⁵ 3,4-Diaminopyridine is a potassium channel blocker, prevents repolarization of the nerve terminal, allows more calcium entry into the cell and increases Ach release.⁶²

CONCLUSION

PNS are uncommon but important clinical conditions. They often precede the identification of the cancer and, if recognized, may lead to earlier tumor discovery and a better chance of cure. Certain onconeural antibodies can be helpful in identifying these conditions, but their role is still being investigated. Treatment of PNS remains a challenge, but therapeutic trials are ongoing.

KEY ILLUSTRATIVE POINTS

- Paraneoplastic neurological syndromes often antedate the diagnosis of cancer, offering an opportunity for detecting cancer at an earlier and curable stage.
- Tests for paraneoplastic antibodies are often negative and do not rule out the diagnosis of a paraneoplastic syndrome.
- Certain clinical presentations should suggest a paraneoplastic syndrome, even in the absence of paraneoplastic antibodies, and prompt a thorough search for occult malignancy.

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